Profit and Loss

(Some useful formulae)

(1.) (G) Gain = Selling price − Cost price (SP − CP)

$(G = SP - CP)$

(2.) $L = CP - SP$

(3.) $G\% = \frac{G}{CP} \times 100$, $L\% = \frac{L}{CP} \times 100$

Q6. Cost price of a toy was Rs 81. It is sold at a profit of 10% on selling price. Find selling price of the toy.

A) Rs.72 B) Rs.99 C) Rs.100 D) Rs.90

$SP = 100$

$G = 10\%$ on $100 = 10$

$CP = 100 - 10 = 90$

CP	SP
90 —	100
1 —	$\frac{100}{90}$
81 —	$\frac{100 \times 81}{90} = 90$ ₹

OR

$81 + \frac{x}{10} = x$

$\Rightarrow \frac{10x - x}{10} = 81$

$\Rightarrow \frac{9x}{10} = 81 \Rightarrow x = \frac{810}{9}$

$= 90$ ₹

Q1. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?

A) 30% B) 70% ✓ C) 100% D) 236%

Let $CP = 100$

$G = 320\%$ of $100 = 320$

$SP = 100 + 320 = 420$

$CP_2 = 125$

$G_2 = 420 - 125 = 295$

$G\%$ w.r.t $SP = \frac{G}{SP} \times 100$

$= \frac{295}{420} \times 100$

$\simeq \frac{\cancel{300}}{\cancel{400}} \times 100 = 75\%$

Ans = 70%

maine → $\frac{295}{125} \times 100$ kara
(125 → CP)
jo ki wrong hai (236% ✗)
SP ke respect me pucha hai to CP lenge niche.

Q3. Joey has 12 eggs with him. He sells x at a profit of 10% and remaining at a loss of 10%. He gains 5% on the whole. What is the value of x?

A) 7 B) 9 ✓ C) 8 D) 10

12 → x (+10%), $12 - x$ (−10%); +5%

$x \times \frac{110}{100} + (12 - x) \times \frac{90}{100} = 12 \times \frac{105}{100}$

$110x + 12 \times 90 - 90x = 12 \times 105$

$20x = 12 \times 105 - 12 \times 90$

$20x = 12(105 - 90) = 12 \times 15$

$x = \frac{\cancel{12}^{3} \times \cancel{15}^{3}}{\cancel{20}_{\cancel{5}}} = 9$

Simple Interest (SI)

(P = Principle, Time = T (in yrs), Rate = R% per annum, A = amount)

$$\boxed{SI\,(I) = \frac{P \times R \times T}{100}} \text{ and } \boxed{A = P + I}$$

(I = interest)

A → final amount

$(\therefore I = A - P)$

Q1. In how many years will a sum double itself at 12.5% p.a. simple interest?
A) 4 B) 8 C) 10 D) 16

Let $P = 100$
$A = 200$
$SI = 100$
$R = 12.5\%$
$SI = \frac{PRT}{100}$

$A = P + I$
$I = A - P$

$T = \frac{SI \times 100}{P \times R}$

$= \frac{\cancel{100} \times 100}{\cancel{100} \times 12.5} = 8$

Q3. A sum of Rs.800 amounts to Rs.920 in 3 years at SI. If the interest rate is increased by 3% it would amount to how much?
A) 992 B) 800 C) 900 D) 920

$P = 800$
$A = 920$
$T = 3y$
$I = 120$
$R = \frac{\cancel{120}^{40} \times 100}{\cancel{800} \times \cancel{3}} = 5\%$

$SI_2 = \frac{800 \times 8 \times 3}{100} = 192$

$A_2 = 800 + 192 = 992$

$Inc = 3\%$ of $800 = 24$
Total $Inc = 24 \times 3 = 72$
$A_2 = 920 + 72$
$= 992$

Q6. A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The sum is:
A. Rs. 650 B. Rs. 690 C. Rs. 698 D. Rs. 700

$A_3 = P + I_3 = 815$ —(1)
$A_4 = P + I_4 = 854$ —(2)
(2) − (1)
$I_4 - I_3 = I_1 = 39$
$I_3 = 39 \times 3 = 117$
$P = A_3 - I_3 = 815 - 117 = 698$

Compound Interest

Formula

If,
Principal = P
Rate = R% per annum
Time = T years

$$A = P\left(1 + \frac{R}{100}\right)^T$$

$$CI = A - P$$

$P = 100$
$T = 3y$
$R = 10\%$
$CI = ?$

100
I ↓ +10% = 10
110
II ↓ +10% = 11 = 10 + 1
121
III ↓ +10% = 12.1 = 10 + 1 + 1.1 = 10 + 1 + 1 + 0.1
133.1

$CI = 133.1 - 100$
$= 33.1$

Q2. The compound interest on Rs.30,000 at 7% p.a. is Rs.4347. The period (in years) is ________.

A) 3 years B) 4 years C) 2 years D) 1 year

4347

30000

I $\downarrow$ +7% = 2100

32100 +

II $\downarrow$ +7% = 2247

= 4347

Q4. A sum of money at compound interest doubled at a certain rate in 4 years. In how many years will it become 8 times at the same rate?

A) 24 B) 12 C) 16 D) 18

$P \xrightarrow{4y} 2P \xrightarrow{4y} 4P \xrightarrow{4y} 8P$

4+4+4 = 12

Q6. What will Rs.2000 amount to in two years if it is invested in 20% p.a. compound interest, interest being compounded semiannually?

A) Rs.2880 B) Rs.3160 C) Rs.2928.20 D) Rs.3148.40

2000

I 6M $\downarrow$ +10% = 200

2200

6M $\downarrow$ +10% = 220

2420

II 6M $\downarrow$ +10% = 242

2662

6M $\downarrow$ +10% = 266.2

2928.2

Imp

It says semiannually not annually